

Lab 2: HTML, CSS, and Responsive Design

Course: CST150: Web technologies

Objective

By the end of this lab, students will:

- Understand the fundamentals of responsive design.
- Learn how to use media queries in CSS.
- Implement a web page with a responsive grid layout.

As always, don't just follow the workshop verbatim, tweak the changes and see the difference. Share your findings with the group.

Pre-requisites

- Completion of Lab 1 (Introduction to Frontend Development & Grid Layouts).
 - Basic knowledge of HTML and CSS.
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Lab Tasks

Task 1: Enhancing the Grid Layout with Responsiveness

1. Duplicate `CST150_lab1` folder into `CST150_lab2` folder.
2. Modify `styles.css` to add media queries for responsiveness.
3. Adjust the grid layout to change on smaller screens.

```
@media (max-width: 768px) {  
  body {  
    grid-template-areas:  
      "header"  
      "nav"  
      "main"  
      "aside"  
      "footer";  
    grid-template-columns: 1fr;  
  }  
  nav { text-align: center; }  
}
```

Task 2: Creating a Responsive Navigation Bar

1. Modify `index.html` to include a basic navigation bar:

```
<nav>
  <ul>
    <li><a href="#">Home</a></li>
    <li><a href="#">About</a></li>
    <li><a href="#">Services</a></li>
    <li><a href="#">Contact</a></li>
  </ul>
</nav>
```

2. Add styles in `styles.css` for better mobile navigation:

```
nav ul {
  list-style: none;
  padding: 0;
  display: flex;
  justify-content: space-around;
}

nav a {
  text-decoration: none;
  color: white;
  padding: 10px;
}

@media (max-width: 600px) {
  nav ul {
    flex-direction: column;
    align-items: center;
  }
}
```

Task 3: Testing Responsiveness

1. Open `index.html` in a browser.
 2. Resize the browser window to observe layout changes.
 3. Test on different devices if possible.
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Discussion Questions

1. How do media queries enhance user experience?
 2. What changes occur in the layout when resizing the window?
 3. How would you improve navigation design for mobile users?
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Bonus Challenge

- Implement a mobile-friendly hamburger menu.
 - Use flexbox alongside grid for improved responsiveness.
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Conclusion

In this lab, students explored responsive design techniques using CSS Grid and media queries. Future lessons will cover advanced styling techniques and integrating JavaScript for interactivity.

End of Lab 2